# MATH 4820 / Regression Analysis


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## Diagnostics for Leverage and Influence


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# 6.1 Importance of Detecting Influential Observations

- **Leverage Point:**
  - unusual x-value;
  - very little effect on regression coefficients.

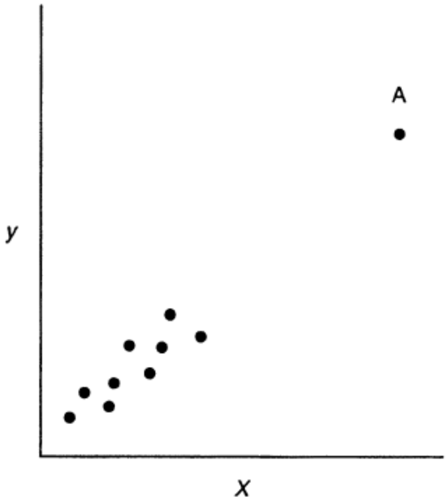


**Figure 6.1** An example of a leverage point.

- **Influence Point:**
  - unusual in y and x;

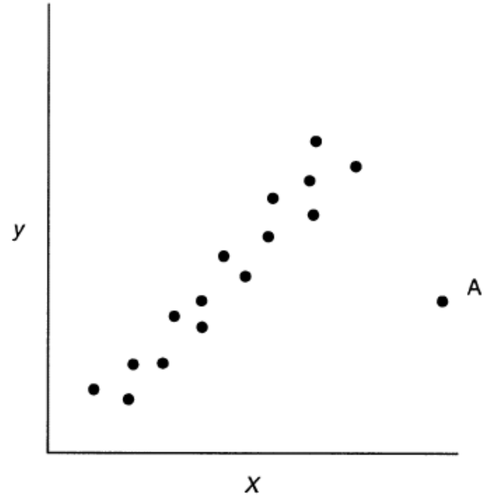


**Figure 6.2** An example of an influential observation.

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# 6.2 Leverage

- The **hat matrix** is:

$$\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$$

- The diagonal elements of the hat matrix are given by

$$h_{ii} = \boldsymbol{x}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_i$$

- $h_{ii}$ – standardized measure of the distance of the *i*th observation from the center of the x-space.
- The average size of the hat diagonal is $p/n$.
- Traditionally, any $h_{ii} > 2p/n$ indicates a **leverage** point.
- An observation with large $h_{ii}$ and a large residual is likely to be **influential**

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# Example 6.1

# The Delivery Time Data

**TABLE 6.1 Statistics for Detecting Influential Observations for the Soft Drink Delivery Time Data**

| Observation $i$ | (a) $h_{ii}$ | (b) $D_i$ | (c) $DFFITS_i$ | (d) Intercept $DFBETAS_{0,i}$ | (e) Cases $DFBETAS_{1,i}$ | (f) Distance $DFBETAS_{2,i}$ | (g) $COVRATIO_i$ |
|---|---|---|---|---|---|---|---|
| 1 | 0.10180 | 0.10009 | −0.5709 | −0.1873 | 0.4113 | −0.4349 | 0.8711 |
| 2 | 0.07070 | 0.00338 | 0.0986 | 0.0898 | −0.0478 | 0.0144 | 1.2149 |
| 3 | 0.09874 | 0.00001 | −0.0052 | −0.0035 | 0.0039 | −0.0028 | 1.2757 |
| 4 | 0.08538 | 0.07766 | 0.5008 | 0.4520 | 0.0883 | −0.2734 | 0.8760 |
| 5 | 0.07501 | 0.00054 | −0.0395 | −0.0317 | −0.0133 | 0.0242 | 1.2396 |
| 6 | 0.04287 | 0.00012 | −0.0188 | −0.0147 | 0.0018 | 0.0011 | 1.1999 |
| 7 | 0.08180 | 0.00217 | 0.0790 | 0.0781 | −0.0223 | −0.0110 | 1.2398 |
| 8 | 0.06373 | 0.00305 | 0.0938 | 0.0712 | 0.0334 | −0.0538 | 1.2056 |
| 9 | 0.49829 | 3.41835 | 4.2961 | −2.5757 | 0.9287 | 1.5076 | 0.3422 |
| 10 | 0.19630 | 0.05385 | 0.3987 | 0.1079 | −0.3382 | 0.3413 | 1.3054 |
| 11 | 0.08613 | 0.01620 | 0.2180 | −0.0343 | 0.0925 | −0.0027 | 1.1717 |
| 12 | 0.11366 | 0.00160 | −0.0677 | −0.0303 | −0.0487 | 0.0540 | 1.2906 |
| 13 | 0.06113 | 0.00229 | 0.0813 | 0.0724 | −0.0356 | 0.0113 | 1.2070 |
| 14 | 0.07824 | 0.00329 | 0.0974 | 0.0495 | −0.0671 | 0.0618 | 1.2277 |
| 15 | 0.04111 | 0.00063 | 0.0426 | 0.0223 | −0.0048 | 0.0068 | 1.1918 |
| 16 | 0.16594 | 0.00329 | −0.0972 | −0.0027 | 0.0644 | −0.0842 | 1.3692 |
| 17 | 0.05943 | 0.00040 | 0.0339 | 0.0289 | 0.0065 | −0.0157 | 1.2192 |
| 18 | 0.09626 | 0.04398 | 0.3653 | 0.2486 | 0.1897 | −0.2724 | 1.0692 |
| 19 | 0.09645 | 0.01192 | 0.1862 | 0.1726 | 0.0236 | −0.0990 | 1.2153 |
| 20 | 0.10169 | 0.13246 | −0.6718 | 0.1680 | −0.2150 | −0.0929 | 0.7598 |
| 21 | 0.16528 | 0.05086 | −0.3885 | −0.1619 | −0.2972 | 0.3364 | 1.2377 |
| 22 | 0.39158 | 0.45106 | −1.1950 | 0.3986 | −1.0254 | 0.5731 | 1.3981 |
| 23 | 0.04126 | 0.02990 | −0.3075 | −0.1599 | 0.0373 | −0.0527 | 0.8897 |
| 24 | 0.12061 | 0.10232 | −0.5711 | −0.1197 | 0.4046 | −0.4654 | 0.9476 |
| 25 | 0.06664 | 0.00011 | −0.0176 | −0.0168 | 0.0008 | 0.0056 | 1.2311 |

# Example 6.1
## The Delivery Time Data

- Examine Table 6.1; if some possibly influential points are removed here is what happens to the coefficient estimates and model statistics:

| Run | $\hat{\beta}_0$ | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $MS_{Res}$ | $R^2$ |
|---|---|---|---|---|---|
| 9 and 22 in | 2.341 | 1.616 | 0.014 | 10.624 | 0.9596 |
| 9 out | 4.447 | 1.498 | 0.010 | 5.905 | 0.9487 |
| 22 out | 1.916 | 1.786 | 0.012 | 10.066 | 0.9564 |
| 9 and 22 out | 4.643 | 1.456 | 0.011 | 6.163 | 0.9072 |

# 6.3 Measures of Influence

- The influence measures discussed here are those that measure the effect of deleting the *i*th observation.

1. Cook's $D_i$, which measures the effect on $\hat{\boldsymbol{\beta}}$
2. $DFBETAS_{j(i)}$, which measures the effect on $\hat{\beta}_j$
3. $DFFITS_i$, which measures the effect on $\hat{Y}_i$
4. $COVRATIO_i$, which measures the effect on the variance-covariance matrix of the parameter estimates.

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# 6.3 MEASURES OF INFLUENCE:

# COOK'S D

$$D_i(\mathbf{M}, c) = \frac{\left(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}}\right)' \mathbf{M} \left(\boldsymbol{\beta}_{(i)} - \hat{\boldsymbol{\beta}}\right)}{c}, \quad i = 1, 2, \ldots, n$$

$$D_i\left(\mathbf{X}'\mathbf{X}, pMS_{\mathrm{Res}}\right) \equiv D_i = \frac{\left(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}}\right)' \mathbf{X}'\mathbf{X} \left(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}}\right)}{pMS_{\mathrm{Res}}}, \quad i = 1, 2, \ldots, n$$

$$D_i = \frac{r_i^2}{p} \frac{\mathrm{Var}(\hat{y}_i)}{\mathrm{Var}(e_i)} = \frac{r_i^2}{p} \frac{h_{ii}}{1 - h_{ii}}, \quad i = 1, 2, \ldots, n$$

What contributes to $D_i$:

1. How well the model fits the *i*th observation, $y_i$
2. How far that point is from the remaining dataset.

Large values of $D_i$ indicate an influential point, usually if $D_i > 1$.

# 6.3 Measures of Influence: Cook's D

**Example 6.2 The Delivery Time Data**

Column b of Table 6.1 contains the values of Cook's distance measure for the soft drink delivery time data. We will illustrate the calculations by considering the first observation. The studentized residual for the delivery time data are in Table 4.1, and $r_1 = -1.6277$. Thus, the value of $D_1$ is

$$D_1 = \frac{r_1^2}{p} \frac{h_{11}}{(1 - h_{11})}$$

$$= \frac{(-1.6277)^2}{3} \frac{0.10180}{(1 - 0.10180)} = 0.10009$$

The largest value of the $D_i$ statistic is $D_9 = 3.41835$, which indicates that deletion of observation 9 would move the least-squares estimate to approximately the boundary of a 96% confidence region around $\hat{\boldsymbol{\beta}}$. The next largest value is $D_{22} = 0.45106$, and deletion of point 22 will move the estimate of $\boldsymbol{\beta}$ to approximately the edge of a 35% confidence region. Therefore, we would conclude that observation 9 is definitely influential using the cutoff of unity, and observation 22 is not influential. Notice that these conclusions agree quite well with those reached in Example 6.1 by examining the hat diagonals and studentized residuals separately.

# 6.4 Measures of Influence: DFFITS and DFBETAS

DFBETAS – measures how much the regression coefficient changes in standard deviation units if the *i*th observation is removed.

$$DFBETAS_{j,i} = \frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{\sqrt{S_{(i)}^2 C_{jj}}}$$

where $\hat{\beta}_{j(i)}$ is an estimate of the *j*th coefficient when the *i*th observation is removed.

- Large DFBETAS indicates *i*th observation has considerable influence. In general, $|DFBETAS_{j,i}| > 2/\sqrt{n}$

# 6.4 Measures of Influence:

# DFFITS and DFBETAS

DFFITS – measures the influence of the *i*th observation on the fitted value, again in standard deviation units.

$$DFFITS_i = \frac{\hat{y}_i - \hat{y}_{(i)}}{\sqrt{S_{(i)}^2 h_{ii}}}$$

- Cutoff: If $|\mathrm{DFFITS}_i| > 2\sqrt{p/n}$, the point is most likely influential.

**Equivalencies**

- See the computational equivalents of both DFBETAS and DFFITS (page 217). You will see that they are both functions of R-student ($t_i$) and $h_{ii}$.

# 6.4 Measures of Influence: DFFITS and DFBETAS

**Example 6.3 The Delivery Time Data**

Columns c to f of Table 6.1 present the values of $DFFITS_i$ and $DFBETAS_{j,i}$ for the soft drink delivery time data. The formal cutoff value for $DFFITS_i$ is $2\sqrt{p/n} = 2\sqrt{3/25} = 0.69$. Inspection of Table 6.1 reveals that both points 9 and 22 have values of $DFFITS_i$ that exceed this value, and additionally $DFFITS_{20}$ is close to the cutoff.

Examining $DFBETAS_{j,i}$ and recalling that the cutoff is $2/\sqrt{25} = 0.40$, we immediately notice that points 9 and 22 have large effects on all three parameters. Point 9 has a very large effect on the intercept and smaller effects on $\hat{\beta}_1$ and $\hat{\beta}_2$, while point 22 has its largest effect on $\hat{\beta}_1$. Several other points produce effects on the coefficients that are close to the formal cutoff, including 1 (on $\hat{\beta}_1$ and $\hat{\beta}_2$), 4 (on $\hat{\beta}_0$), and 24 (on $\hat{\beta}_1$ and $\hat{\beta}_2$). These points produce relatively small changes in comparison to point 9.

Adopting a diagnostic view, point 9 is clearly influential, since its deletion results in a displacement of every regression coefficient by at least 0.9 standard deviation. The effect of point 22 is much smaller. Furthermore, deleting point 9 displaces the predicted response by over four standard deviations. Once again, we have a clear signal that observation 9 is influential.

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# 6.5 A Measure of Model Performance

- Generalized variance of $\hat{\boldsymbol{\beta}}$ as

$$\mathrm{GV}\left(\hat{\boldsymbol{\beta}}\right) = \left|\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right)\right| = \left|\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}\right|$$

- Information about the overall precision of estimation can be obtained through another statistic, $\mathrm{COVRATIO}_i$

$$COVRATIO_i = \frac{|(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1} S^2_{(i)}|}{|(\mathbf{X}'\mathbf{X})^{-1} MS_{\mathrm{Res}}|}$$

$$= \frac{(S^2_{(i)})^p}{MS^p_{\mathrm{Res}}}\left(\frac{1}{1-h_{ii}}\right)$$

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# 6.5 A Measure of Model Performance

**Cutoffs and Interpretation**

- If $COVRATIO_i > 1$, the *i*th observation improves the precision.
- If $COVRATIO_i < 1$, *i*th observation can degrade the precision.

Or,

- Cutoffs:
  $COVRATIO_i > 1 + 3p/n$ or $COVRATIO_i < 1 - 3p/n$;
  (the lower limit is really only good if n > 3p).

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# 6.5 A Measure of Model Performance

**Example 6.4 The Delivery Time Data**

Column g of Table 6.1 contains the values of $COVRATIO_i$ for the soft drink delivery time data. The formal recommended cutoff for $COVRATIO_i$ is $1 \pm 3p/n = 1 \pm 3(3)/25$, or 0.64 and 1.36. Note that the values of $COVRATIO_9$ and $COVRATIO_{22}$ exceed these limits, indicating that these points are influential. Since $COVRATIO_9 < 1$, this observation degrades precision of estimation, while since $COVRATIO_{22} > 1$, this observation tends to improve the precision. However, point 22 barely exceeds its cutoff, so the influence of this observation, from a practical viewpoint, is fairly small. Point 9 is much more clearly influential.

# 6.6 Detecting Groups of Influential Observations

- Previous diagnostics were "single-observation"
- It is possible that a **group** of points have high-leverage or exert undue influence on the regression model.
- Multiple-observation deletion diagnostic can be implemented.
- Cook's D can be extended to incorporate **multiple** observations:

$$D_{\mathbf{i}}(\mathbf{X}'\mathbf{X}, pMS_{\mathrm{Res}}) = D_{\mathbf{i}} = \frac{(\hat{\boldsymbol{\beta}}_{(\mathbf{i})} - \hat{\boldsymbol{\beta}})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}}_{(\mathbf{i})} - \hat{\boldsymbol{\beta}})}{pMS_{\mathrm{Res}}}$$

where **i** denotes the $m \times 1$ vector of indices specifying the points to be deleted.

# 6.7 Treatment of Influential Observations

- Should an influential point be discarded?

  Yes, if
  - there is an error in recording a measured value;
  - the sample point is invalid; or,
  - the observation is not part of the population that was intended to be sampled

  No, if
  - the influential point is a valid observation.
- Robust estimation techniques
  - These techniques offer an alternative to deleting an influential observation.
  - Observations are retained but ***downweighted*** in proportion to residual magnitude or influence.

# QUESTIONS?

- ANY QUESTION?